

Passive DS-TWR Speed-Limit Study

UWB Module ESP-IDF field validation

27 July 2026

1 Purpose and method

This experiment evaluates how far the custom receive-only Passive DS-TWR protocol can be accelerated while retaining static accuracy. Four anchors formed a surveyed $3\text{ m} \times 3\text{ m}$ square with $\pm 2\text{ mm}$ placement tolerance. The tag was stationary at the surveyed centre, $(1.5, 1.5)\text{ m}$.

All measurements used firmware `137c2b5`. Positions were captured from the lossless dashboard event stream rather than from periodic snapshots. Raw AlgMin estimates and constant-velocity EKF estimates were retained separately. The reported update rate counts complete position frames that contain newly acquired UWB information.

2 Results

Profile	pos/s	nominal pos/s	EKF RMSE cm	EKF P95 cm	raw RMSE cm	success %
Robust 16 ms	49.3	62.5	1.73	2.90	2.05	98.2
Robust 13 ms	50.6	76.9	1.89	3.19	2.21	93.8
Robust 10 ms	61.5	100.0	1.95	3.23	2.25	92.1
Robust 10 ms, 0.9/0.9 ms	62.8	100.0	2.58	4.14	2.82	91.1
Fast Star 10 ms	56.7	100.0	3.83	6.15	4.12	92.9

Table 1: Controlled static measurements. Success is the aggregate anchor-dialogue success rate reported by the firmware.

3 Interpretation

The best fast operating point is **Robust Rotating with a configured 10 ms frame and 1.0/1.0 ms RESP/FINAL delays**. It delivered **61.5 positions/s**, **1.95 cm 2-D RMSE**, and **3.23 cm P95**. Robust 16 ms remains the most accurate profile at 1.73 cm RMSE, but the 10 ms profile adds 24.8% update throughput for only 0.23 cm additional RMSE.

Fast Star did not improve speed in this geometry. Its asymmetric initiator load produced 56.7 positions/s and 3.83 cm RMSE, so Robust Rotating is the better optimization base.

The 0.9/0.9 ms delay experiment was also rejected. It improved throughput by only 2.1%, while RMSE rose from 1.95 to 2.58 cm and P95 from 3.23 to 4.14 cm. The live modules were therefore restored to the 1.0/1.0 ms control timing.

4 Why 85 positions/s was not reached

The nominal 10 ms frame implies 300 directed slots/s. Anchor logs measured approximately 196.7 slots/s, or 5.08 ms per effective slot. A complete independent position frame uses three directed

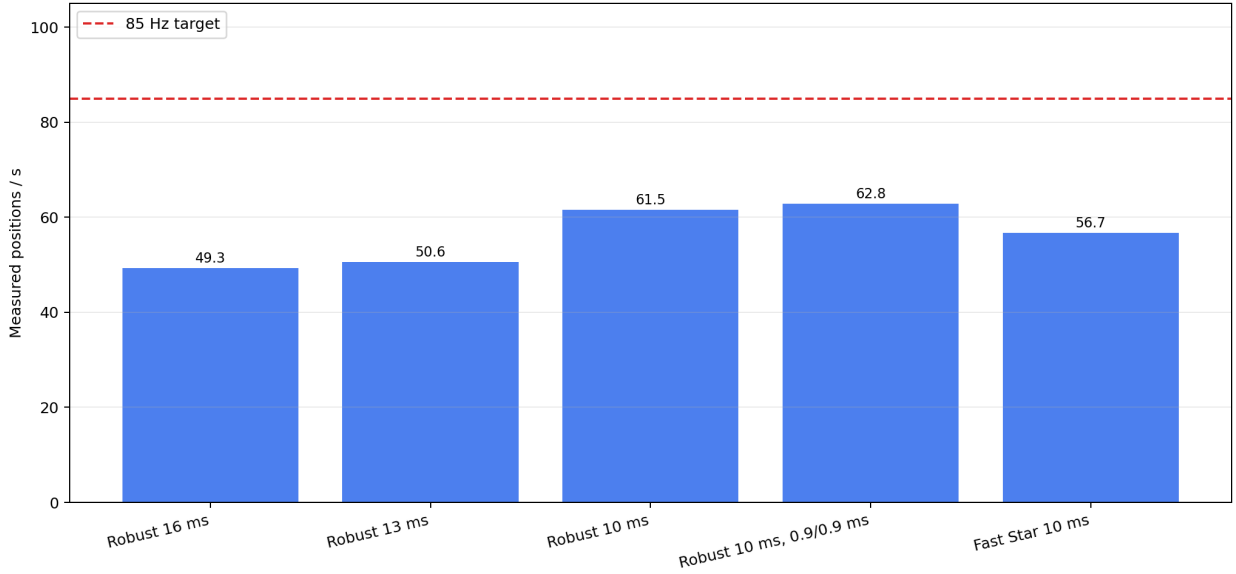


Figure 1: Measured position throughput. The dashed line marks the 85 positions/s target.

anchor dialogues, so 85 positions/s requires at least 255 successful slots/s before allowing for rejections. Accepted position throughput must therefore improve by approximately 38.2%.

The present bottleneck is the radio transaction and scheduling path rather than the dashboard or EKF. Re-solving after every observation could display more than 85 updates/s, but consecutive results would reuse nearly all the same measurements. Such a mode may reduce visual latency, but it must be reported separately from independent frame throughput.

5 Recommended next phase

1. Instrument POLL TX, delayed RESP TX, FINAL TX/RX, CIA readout, and RX re-arm with per-stage timestamps and failure counters.
2. Replace millisecond receive-loop scheduling with DW3000 delayed-TX and delayed-RX deadlines in a non-blocking slot state machine.
3. Retain Robust Rotating at 1.0/1.0 ms as the control. Accept each timing change only if RMSE and P95 remain inside the control confidence band.
4. Evaluate a rolling solve on every new observation as a distinct low-latency display mode, exposing both solver updates/s and independent frames/s.

6 Reproducibility

The repository contains the raw JSONL event streams, status snapshots, firmware timing logs, machine-readable metrics, and the analysis script `tools/uwb_passive_speed_analyze.py`.

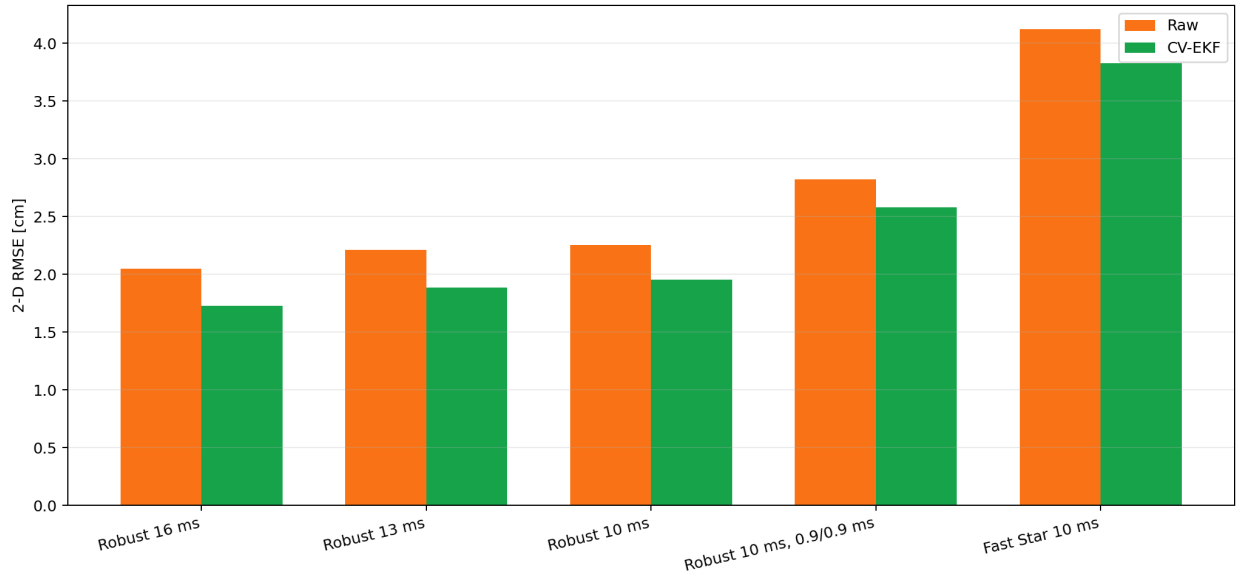


Figure 2: The CV-EKF reduces dispersion at every tested operating point.

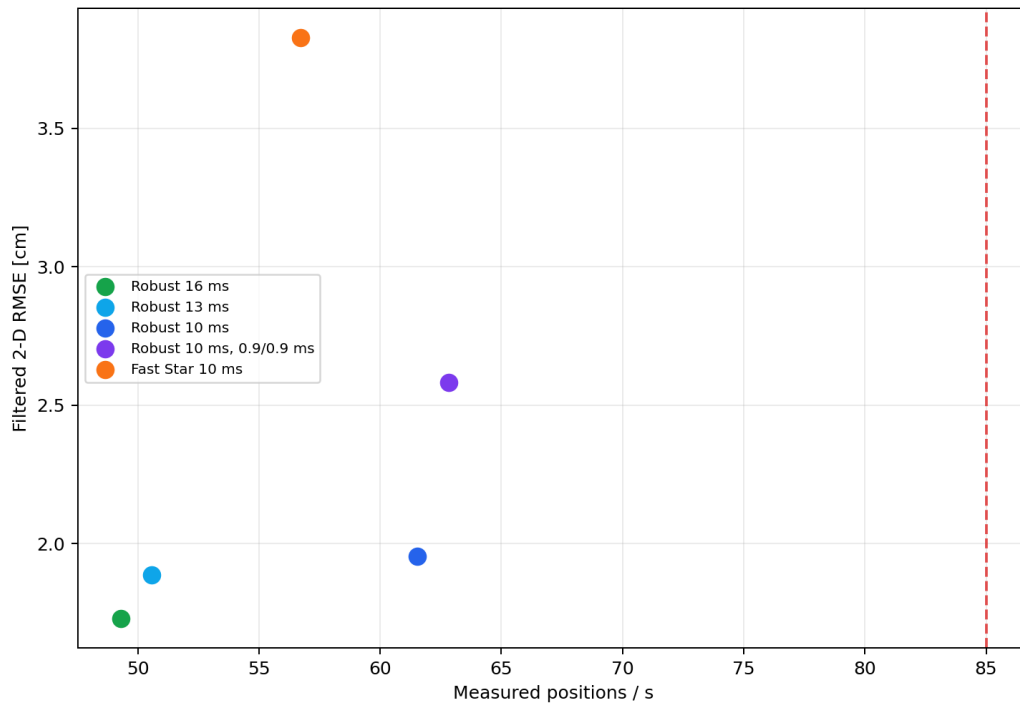


Figure 3: Speed-accuracy trade-off for the tested profiles.